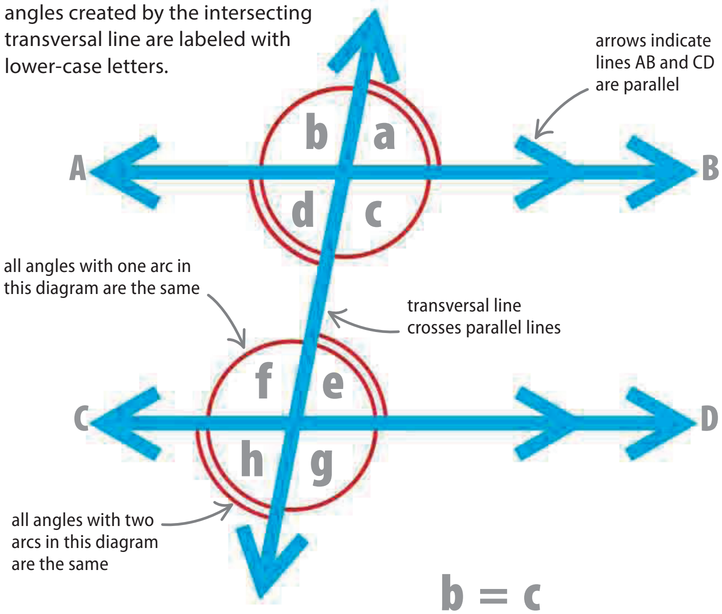
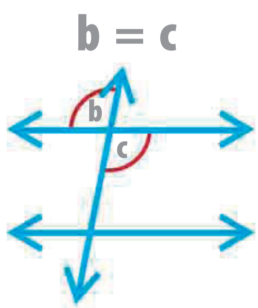
Angles and parallel lines

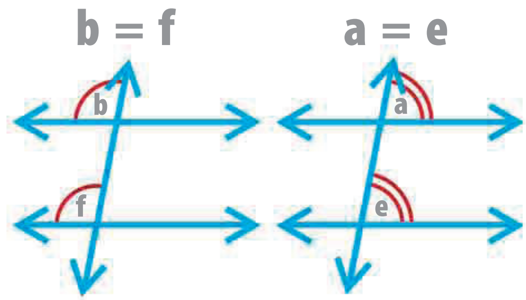
Angles can be grouped and named according to their relationships with straight lines. When parallel lines are crossed by a transversal, it creates pairs of equal angles—each pair has a different name.

▽ Labeling angles
Lines AB and CD are parallel. The angles created by the intersecting transversal line are labeled with lower-case letters.

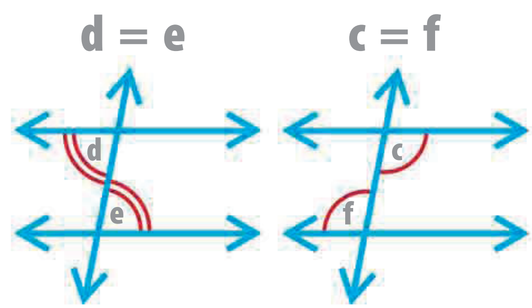




▷ Vertical angles
When two lines cross, equal angles are formed on opposite sides of the point. These angles are known as vertical angles.



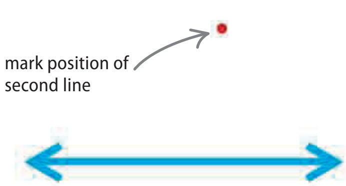
△ Corresponding angles
Angles in the same position in relation to the transversal line and one of a pair of parallel lines, are called corresponding angles. These angles are equal.



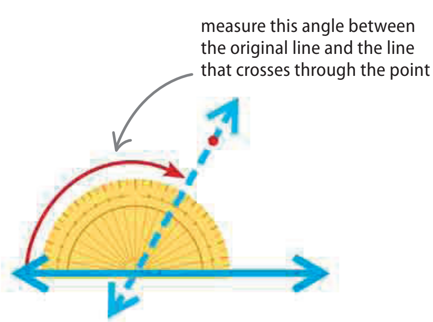
△ Alternate angles
Alternate angles are formed on either side of a transversal between parallel lines. These angles are equal.

Drawing a parallel line

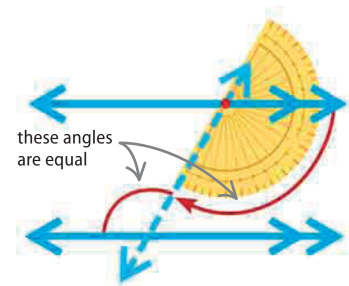
Drawing a line that is parallel to an existing line requires a pencil, a ruler, and a protractor.



Draw a straight line with a ruler. Mark a point—this will be the distance of the new, parallel line from the original line.



Draw a line through the mark, intersecting the original line. This is the transversal. Measure the angle it makes with the original line.



Measure the same angle from the transversal. Draw the new line through the mark with a ruler; this line is parallel to the original line.